MILITARY SPECIFICATION

PASTE, COPPER BRAZING, WATER THINNING

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

of steels, copper, copper alloys, and nickel alloys.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

O-E-771	Ethylene Glycol, Inhibited
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Boxes, Fiber
PPP-C-96	Cans, Metal, 28 Gage and Lighter
RR-S-366	Sieves, Standard for Testing Purposes

Military

MIL-B-10377	Box, Wood, Cleated, Veneer, Paper Overlaid

STANDARDS

Federal

Federal Test Method Std. No. 151	Metals; Test Methods

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The materials employed shall produce a copper brazing paste meeting the requirements of this specification.

3.2 Composition of product (by weight).- The composition of the brazing paste shall be as follows:

Active ingredients	-	70 percent minimum.
Vehicle	-	29 percent maximum.
Emulsifying agent	-	1 percent maximum.

3.2.1 Active ingredients.- The active ingredient shall be deoxidized copper powder of the following composition:

Copper	-	99 percent minimum.
Iron	-	0.15 percent maximum.
Other	-	0.85 percent maximum.

3.2.1.1 Granulation of powder.- The copper powder employed, shall be of such fineness that not more than 0.5 percent by weight shall remain on a No. 325 screen conforming to the requirements of Specification RR-S-366.

3.2.2 Vehicle.- The vehicle shall consist of approximately equal proportions by weight of water, and ethylene glycol conforming to Specification O-E-771.

3.2.3 Emulsifying agent.- The emulsifying agent shall consist of Algin or petroleum sulfonate or a mixture thereof.

3.3 Brazing characteristics.- When the brazing paste is applied to a steel test panel, heated to 2,050° ±25°F and held at heat for 30 minutes in a hydrogen atmosphere having a dew point not higher than -40°F, and cooled to 1,000°F or lower in a protective atmosphere, the active ingredient shall be reduced completely to metallic copper and shall melt and flow freely. The vehicle and emulsifying agent shall burn off leaving no carbonaceous residue, and without causing carburization of the panel.

3.4 Workmanship.- The paste shall be an intimately blended mixture of uniform consistency. It shall not separate in the container to such an extent that it cannot be returned to uniform consistency by stirring. Paste diluted with water or ethylene glycol as required, shall have acceptable application requirements.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- All the tests required herein for testing of copper brazing paste are classified as Acceptance Tests, for which necessary sampling techniques and methods of testing are specified in this section.

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4.2 Sampling.-

4.2.1 Sampling for Visual examination.- A random sample shall be selected from each Inspection lot in accordance with Standard MIL-STD-105, AQL of 2.5.

4.2.2 Sampling for Chemical and Usability Tests.- A random sample shall be selected from each Inspection lot in accordance with the appendix to Standard MIL-STD-105, Inspection Level L-5, Acceptance number zero. Preparation of sample specimens for the Chemical analysis test specified in 4.5.2 shall be in conformance with the requirements of Federal Test Method Standard No. 151. For the Usability test specified in 4.5.3, a sufficient amount of material shall be employed to conduct the test.

4.3 Lot.- An inspection lot shall consist of all copper brazing paste of the same composition produced under essentially the same conditions, and submitted for inspection at one time. The unit of inspection shall be 1 pound.

4.4 Report of tests.- Unless otherwise specified, the contractor shall furnish a certificate stating that the material being provided conforms to the requirements of this specification. The test report shall be furnished in duplicate to the procuring activity and shall be signed by the director, or an authorized assistant, of the laboratory making the tests.

4.5 Tests.-

4.5.1 Visual examination.- Inspection shall be made to assure compliance with the workmanship requirements.

4.5.2 Chemical analysis.- Check chemical analysis shall be made by wet chemical, spectrochemical, or other analytical methods. In the event of dispute, analysis shall be made by wet chemical methods. Check analysis may be waived at the discretion of the Government Inspector, provided the material can be positively identified as being part of a lot previously analyzed and found to be in conformance with the chemical composition requirements specified herein, or that the manufacturer's method of composition control is acceptable to the Government Inspector.

4.5.3 Usability test.- The copper brazing paste shall demonstrate satisfactory usability and brazing characteristics as required by 3.3.

4.6 Rejection and retest.- Paragraph titled "Resubmitted lots", of Standard MIL-STD-105 shall apply except that a resubmitted inspection lot shall be inspected [illegible] the supervision of the Government Inspector using tightened

lot is resubmitted, full particulars concerning the cause of previous rejection and the action taken to correct the defects found in the inspection lot shall be furnished by the contractor to the Government Inspector.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.-

5.1.1 Level A.- Unless otherwise specified in the contract or purchase order, copper brazing paste shall be packaged in Type V, Class 1; Type V, Class 2; or Type VI unit containers in accordance with Specification PPP-C-96. The size of the container shall be specified by the procuring activity. (See 6.2.)

5.1.2 Level C.- Packaging shall conform to the manufacturer's commercial practice, unless the procuring activity determines a modification to the manufacturer's commercial practice is required to provide adequate protection.

5.2 Levels of packing.-

5.2.1 General.- The levels of packing to be used shall be specified by the procuring activity (see 6.2). As far as practicable, exterior shipping containers shall contain identical quantities. The gross weight of each shipping container when packed for shipment shall not exceed, for Navy procurement, approximately 200 pounds for both domestic and overseas shipment; and for Air Force procurement, approximately 500 pounds for domestic and 200 pounds for overseas shipments.

5.2.2 Level A.- Exterior packing shall be in accordance with overseas-type containers conforming to Specifications MIL-B-10377, PPP-B-601, and PPP-B-636.

5.2.3 Level C. Exterior packing shall be in accordance with good commercial practice. The shipment shall conform with carrier's applicable rules and regulations at the time of shipment.

5.3 Marking for shipment.- In addition to any other marking specified in the contract or order, unit containers and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- The copper brazing paste specified herein may be used for brazing in accordance with Specification MIL-B-7883, when the use of inserts of copper wire or sheet is impractical and where carburization of the metals being joined is undesirable.

6.2 Ordering data.- Procuring documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether 1-pound or 5-pound containers are desired (see 5.1.1).
(c) Levels of packaging and packing required (see 5.1 and 5.2).

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Air Force

Preparing activity:
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